Supplementary Information

Cross-model uncertainty-aware minimal editing of cis-regulatory elements for cell-type-selective design

Angran Xia
School of Chemistry and Chemical Engineering, Shaanxi Normal University,
620 West Chang'an Avenue, Chang'an District, Xi'an, Shaanxi 710119, China

Correspondence: haangyeon@snnu.edu.cn

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1 Supplementary methods and audit notes

1.1 Study phases and separation of roles

The study was executed in sequential phases. G0–G3 established the data audit, published-model reproduction, pilot CNN reviewer, and initial editing workflow. G4 froze a 90-parent, three-method hard-audit benchmark (six nominal parents excluded for overlap with the G4 exploratory pilot). G6 assigned computational candidate tiers. G7 trained the full-data residual-dilated reviewer and the multi-kernel post-freeze evaluator. G8 selected 600 new parents and compared four search methods. G9 reran seven complete search configurations on 180 parents. G10 applied the post-freeze endpoint and assembled summary analyses. Activity labels were never used to select parents or candidate sequences.

The terms *reviewer* and *post-freeze evaluator* denote different roles. The reviewer was available to SafeEdit during G8/G9 search. The post-freeze evaluator was loaded only after candidate files were frozen. Both were trained on the same public MPRA source table; hence "post-freeze" describes procedural separation from search, not external biological independence.

1.2 Integrity checks

The final G9 table contains 15,120 rows, 180 unique parents, seven configurations, three target cells, and four budgets. There are no duplicate parent–target–budget–configuration keys, no non-finite numeric values, and no exact edit-budget violations among feasible designs. The G8 collapsed table contains 28,800 method-level rows and 600 parents. Random metrics are means of three independent replicates; the collapsed table must not be used to select a particular DNA sequence.

All submission payloads were rechecked after manuscript generation. The release manifest contains SHA-256 checksums for every payload file and excludes the manifest itself from self-hashing.

2 Supplementary results

2.1 Data partitions and predictor performance

Table 1: Table S1. Quality-filtered full-data partitions used for G7 model training.

Partition	Records	Chromosomes	Use
Training	627,661	Eligible chromosomes excluding validation/test	Parameter fitting
Validation	58,811	19, 21, X	Checkpoint choice and variance scaling
Test	62,582	7, 13	Final predictive diagnostics
Total	749,054	–	After quality filtering and exact deduplication

Table 2: Table S2. Ensemble performance on the 62,582-record test split, recomputed from saved candidate-independent prediction tables.

Model	Cell	Pearson r	Spearman ρ	MAE	Uncertainty–error ρ
Reviewer	K562	0.845	0.781	0.396	0.448
Reviewer	HepG2	0.855	0.809	0.354	0.404
Reviewer	SK-N-SH	0.847	0.810	0.390	0.399
Post-freeze evaluator	K562	0.785	0.719	0.463	0.467
Post-freeze evaluator	HepG2	0.795	0.743	0.417	0.416
Post-freeze evaluator	SK-N-SH	0.799	0.760	0.444	0.400

Table 3: Table S3. Validation-scaled interval coverage on the test split.

Model	Cell	80% interval	90% interval	95% interval
Reviewer	K562	0.815	0.897	0.935
Reviewer	HepG2	0.794	0.885	0.930
Reviewer	SK-N-SH	0.814	0.900	0.940
Post-freeze evaluator	K562	0.821	0.900	0.935
Post-freeze evaluator	HepG2	0.789	0.879	0.923
Post-freeze evaluator	SK-N-SH	0.816	0.899	0.935

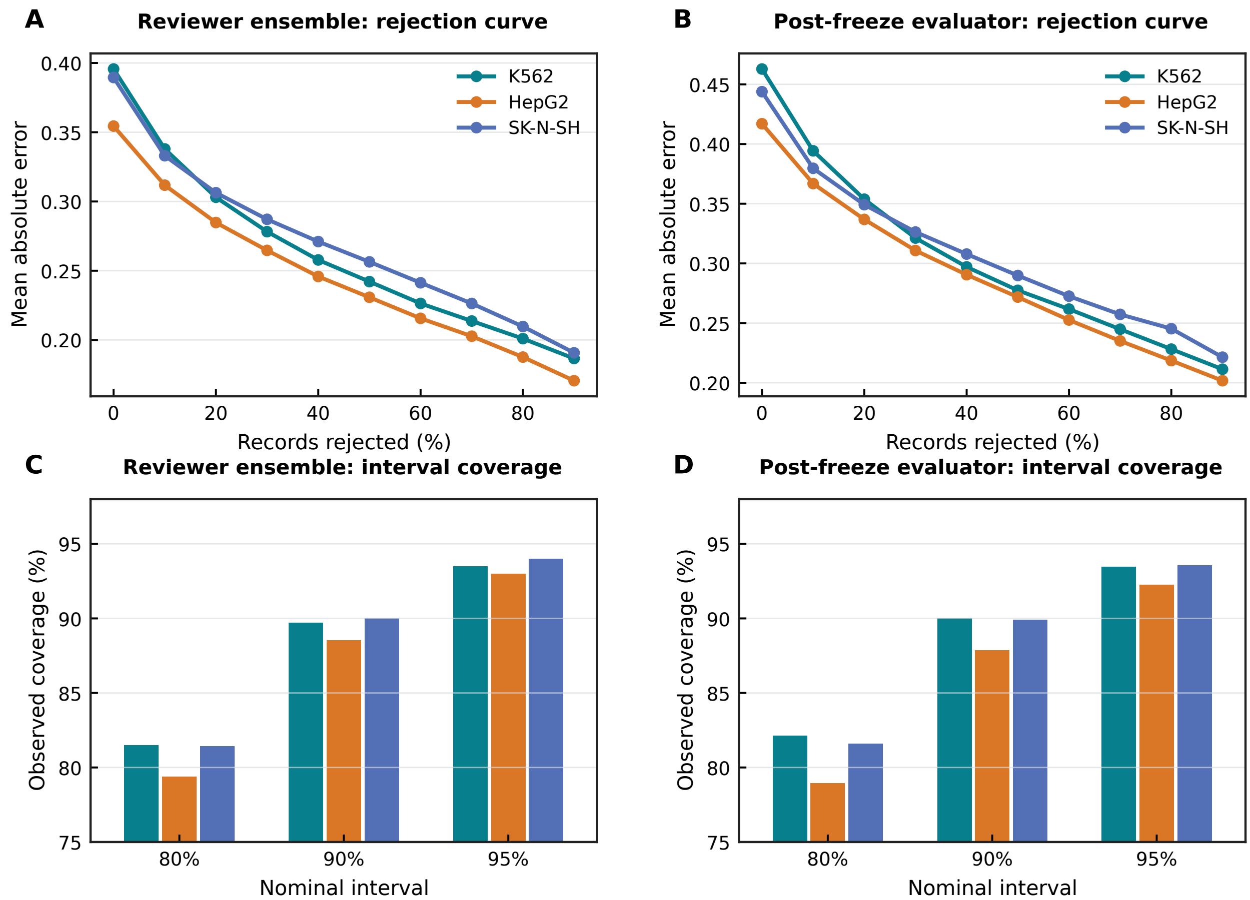

Figure 1: Figure S1. Uncertainty rejection and interval calibration. A–B, mean absolute error for the reviewer and post-freeze ensembles as records with the largest predicted uncertainty are removed from the held-out test set. **C–D**, observed test coverage at 80%, 90%, and 95% nominal levels by cell type. Curves and bars are descriptive test diagnostics; variance scales were fixed from validation only.

2.2 G4 hard-audit stratification

Table 4: Table S4. G4 audit pass rates by target cell and edit budget. Each row contains 90 non-overlapping parents per method after excluding six parents that overlapped the G4 exploratory pilot.

Target	Budget	Random (%)	Greedy (%)	SafeEdit (%)	SafeEdit–Greedy (pp)
K562	1	14.4	58.9	60.0	1.1
K562	5	8.9	56.7	77.8	21.1
K562	10	17.8	43.3	82.2	38.9
K562	20	24.4	50.0	81.1	31.1
HepG2	1	8.9	34.4	35.6	1.1
HepG2	5	5.6	35.6	45.6	10.0
HepG2	10	2.2	33.3	56.7	23.3
HepG2	20	8.9	27.8	60.0	32.2
SK-N-SH	1	24.4	36.7	38.9	2.2
SK-N-SH	5	17.8	35.6	57.8	22.2
SK-N-SH	10	15.6	22.2	64.4	42.2
SK-N-SH	20	8.9	31.1	66.7	35.6

2.3 G8 post-freeze endpoint and heterogeneity

Table 5: Table S5. Pooled G8 method summaries. Each method has 7,200 parent–target–budget records after averaging the three random replicates.

Method	Margin gain	Target gain	Positive margin (%)	Uncertainty
Random matched	0.001	0.012	48.6	0.185
Greedy Malinois	0.822	1.803	97.1	0.329
Primary beam	0.881	1.866	94.8	0.335
SafeEdit consensus	0.877	1.661	95.4	0.314

Table 6: Table S6. Pooled paired G8 contrasts. Differences are SafeEdit minus comparator; confidence intervals use 100,000 parent-cluster bootstrap replicates.

Comparator	Metric	Mean difference	95% CI
Greedy	Post-freeze margin gain	0.0555	0.0398 to 0.0705
Greedy	Post-freeze target gain	-0.1412	-0.1774 to -0.1057
Greedy	Post-freeze uncertainty	-0.0153	-0.0214 to -0.0091
Primary beam	Post-freeze margin gain	-0.0036	-0.0114 to 0.0042
Primary beam	Post-freeze target gain	-0.2048	-0.2294 to -0.1808
Primary beam	Post-freeze uncertainty	-0.0209	-0.0251 to -0.0166
Random matched	Post-freeze margin gain	0.8763	0.8562 to 0.8960
Random matched	Post-freeze target gain	1.6494	1.6034 to 1.6950
Random matched	Post-freeze uncertainty	0.1285	0.1222 to 0.1348

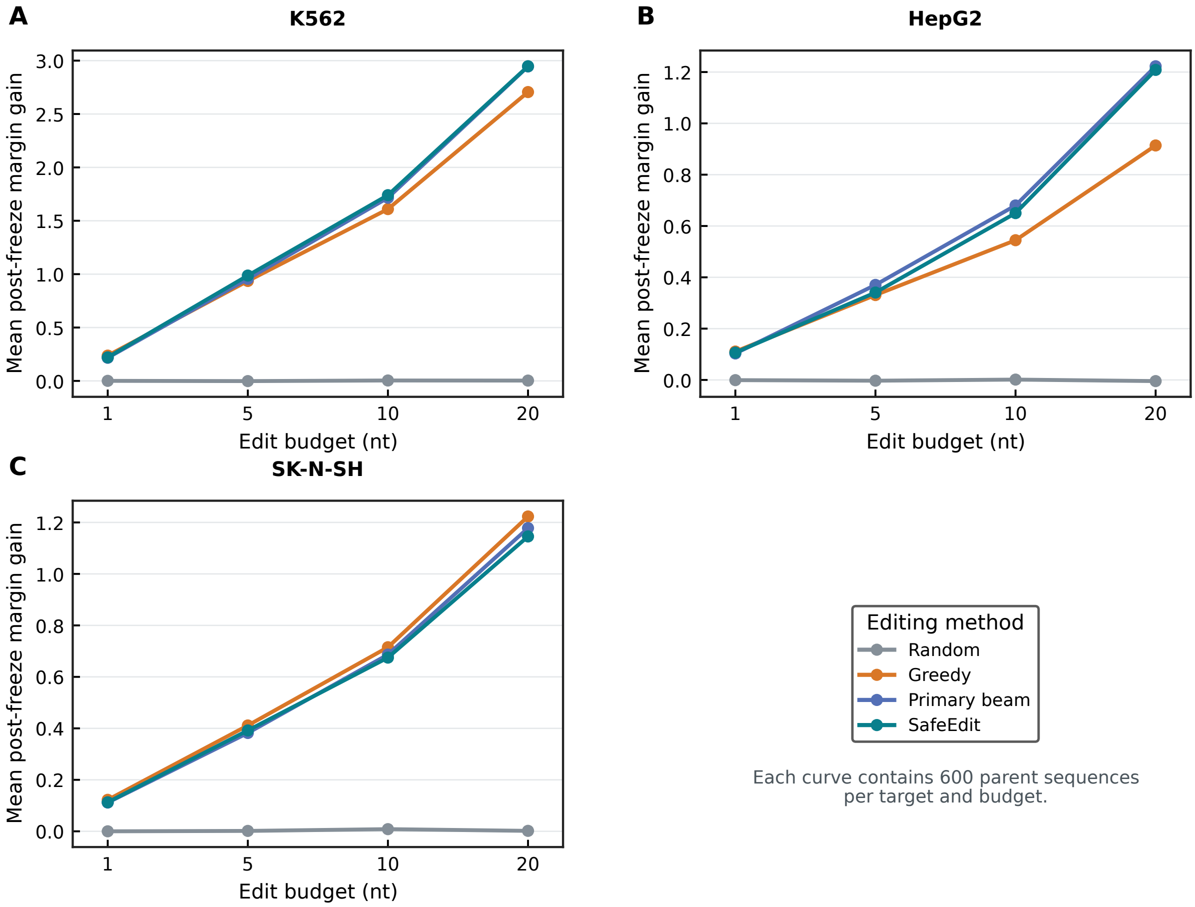

Figure 2: **Figure S2. Target–budget heterogeneity in the G8 post-freeze endpoint. A–C**, mean post-freeze specificity-margin gain across edit budgets for random, greedy, primary-beam, and SafeEdit searches in K562, HepG2, and SK-N-SH. Each point summarizes 600 parents. Values are descriptive and are not multiplicity-adjusted.

2.4 G9 true-search ablation

Table 7: Table S7. G9 post-freeze outcomes for seven independently rerun search configurations. Each configuration contains 2,160 rows from 180 parents, three targets, and four budgets.

Configuration	Margin gain	Target gain	Positive (%)	Uncertainty	Infeasible
Primary beam	0.890	1.897	94.7	0.334	48
SafeEdit full	0.890	1.705	95.3	0.312	48
No reviewer score	0.890	1.897	94.7	0.334	48
No uncertainty penalty	0.891	1.719	95.2	0.313	48
No strand penalty	0.887	1.701	95.4	0.310	48
No naturalness rerank	0.979	2.017	95.3	0.343	48
No sequence prefilter	0.910	1.763	98.1	0.318	0

Table 8: Table S8. Selected G9 paired contrasts. Differences are full SafeEdit minus the ablation; intervals use 100,000 parent-cluster bootstrap replicates.

Comparator	Metric	Mean difference	95% CI
Primary beam	Post-freeze margin gain	0.0002	-0.0145 to 0.0151
Primary beam	Post-freeze target gain	-0.1920	-0.2440 to -0.1432
Primary beam	Post-freeze uncertainty	-0.0212	-0.0283 to -0.0140
No naturalness	Post-freeze margin gain	-0.0888	-0.1022 to -0.0754
No naturalness	Post-freeze target gain	-0.3117	-0.3600 to -0.2658
No naturalness	Post-freeze uncertainty	-0.0305	-0.0394 to -0.0218
No prefilter	Post-freeze margin gain	-0.0198	-0.0387 to -0.0043
No prefilter	Post-freeze target gain	-0.0581	-0.1021 to -0.0227
No prefilter	Post-freeze uncertainty	-0.0060	-0.0102 to -0.0024

2.5 Candidate library

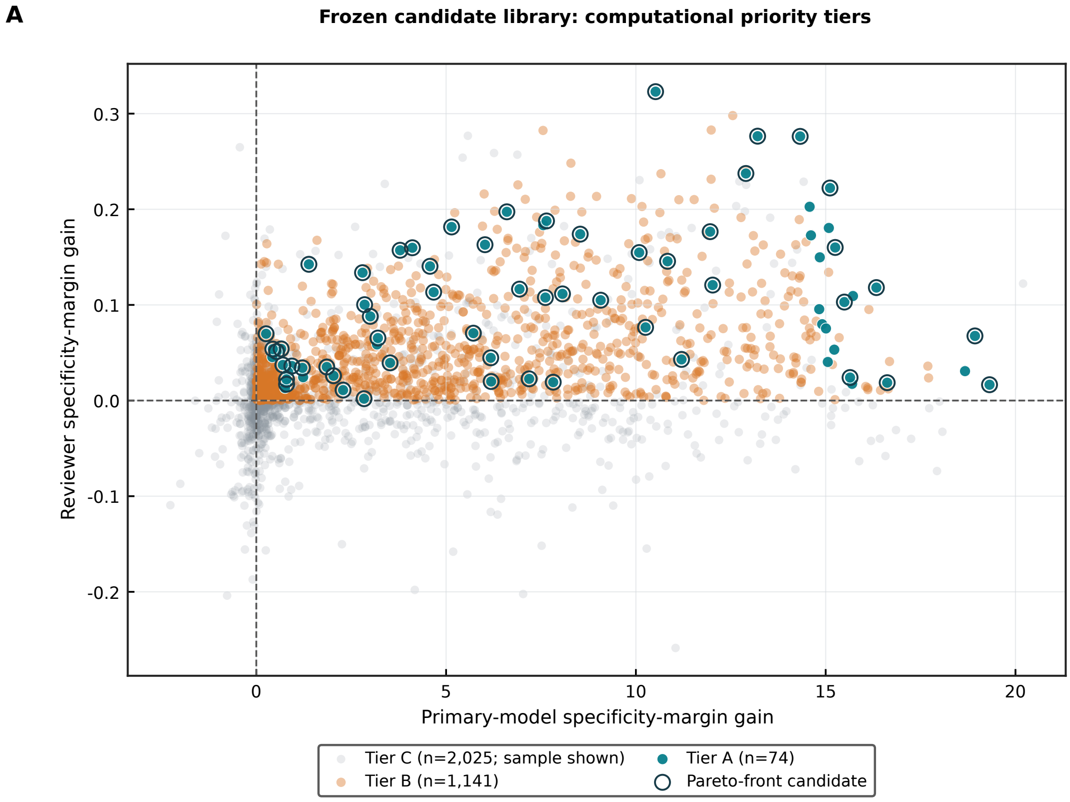

Figure 3: Figure S3. Computational candidate tiers. Primary-model and reviewer specificity-margin gains are shown for the frozen 3,240-sequence G4 library. All Tier A and Tier B candidates are plotted; Tier C is downsampled for legibility. Outlined points lie on the computed Pareto frontier. Tier labels represent computational priorities for experimental screening.

Table 9: Table S9. Frozen G4 candidate tiers.

Tier	Candidates	Fraction (%)	Intended use
A	74	2.3	Highest-priority experimental panel
B	1,141	35.2	Alternative computational candidates
C	2,025	62.5	Method and failure-mode analysis

2.6 Representative Tier A candidates

Table 10: Table S10. Three representative Tier A reporter-screening candidates, one per cell type, selected from the locked non-overlapping 90-parent nine-criterion design benchmark. Representative examples were selected according to a prespecified rule requiring Tier A status, one example per target cell, positive cross-model margin gain, reduced uncertainty, and no synthesis-risk flag. Margins are shown for the primary Malinois predictor and the held-out reviewer ensemble; all three achieved Pareto-frontier status and carry no synthesis-risk flags. Edit positions use 0-indexed coordinates within the 200-nt sequence window. Margin and uncertainty values are in log-fold-change units. GC is expressed as a fraction. All edits are substitutions within a budget of five nucleotides. Motif-level observations at edited positions are reported as testable sequence-grammar hypotheses for follow-up.

Property	K562 (chr7:127284707)		HepG2 (chr7:105701659)		SK-N-SH (chr13:23368564)	
	Parent	Edited	Parent	Edited	Parent	Edited
Edit budget	5		5		5	
Edit positions	89, 118, 130, 132, 180		89, 94, 99, 112, 199		1, 148, 149, 171, 174	
Edit details	89T>A; 118C>G; 130A>T; 132C>A; 180C>A		89G>T; 94C>A; 99G>C; 112G>C; 199C>G		1C>G; 148T>C; 149C>T; 171A>G; 174C>A	
Primary margin	-0.372	$+4.204$	-0.620	$+5.100$	$+0.020$	$+2.874$
Reviewer margin	-0.138	$+0.002$	-0.068	$+0.003$	-0.197	-0.097
Primary margin gain	$+4.577$		$+5.719$		$+2.854$	
Reviewer margin gain	$+0.140$		$+0.070$		$+0.100$	
Primary target gain	$+4.558$		$+6.093$		$+3.427$	
Reviewer uncertainty	1.253	1.099	1.243	1.143	0.552	0.445
Uncertainty change	-0.154		-0.100		-0.107	
GC content	0.575	0.565	0.570	0.560	0.365	0.365
Naturalness Δ	$+0.056$		$+0.006$		$+0.028$	
Max homopolymer	4	4	5	5	5	5
Synthesis risk	None		None		None	
Pareto frontier	Yes		Yes		Yes	
Parent sequence	TGAGGGCAGAGCTCAGCCGCAACCCGGCCCTGCTGCTTAGAGGCTCGGGGCCTTGGCAAATGACAGCCTCCCTGAGCCTGTTTCTCCATCCGTAAAACAGGCCTCCCAGCCCTGTGTCGGAGCAACTGAAACGAGGGTTAAATGAGGAAACCATGCGACAGATGTCAGCATAATCCCTGCCACGCAGAGGAGGCAAAATC					
Edited sequence (K562)	TGAGGGCAGAGCTCAGCCGCAACCCGGCCCTGCTGCTTAGAGGCTCGGGGCCTTGGCAAATGACAGCCTCCCTGAGCCTGTTTCTCCAACCGTAAAACAGGCCTCCCAGCCCTGTGTGGGAGCAACTGATAAGAGGGTTAAATGAGGAAACCATGCGACAGATGTCAGCATAATCCCTGACACGCAGAGGAGGCAAAATC					
Parent sequence (HepG2)	GGGGGTGAGGGATCCCTGCAATACATAACACGTGCTAAGCAGTGGGGTTTGGGTGCTACTTCCCAACTGGCCGCTCTTAGGAACTGGTGAAACCTTAAGTCAGGTGTGACTGAGCGAGCCCCTGGGTCATGGCTATTCTCCTTGCTGCTCTGCTCTCCTAAGTCTGGTTTGAGGCTCCTCTCCCTCCCTGGCGTCCCACC					

Property	K562 P	K562 E	HepG2 P	HepG2 E	SK-N-SH P	SK-N-SH E
Edited sequence (HepG2)	GGGGGTGAGGGATCCCTGCAATACATAACACGTGCTAAGCAGTGGGGTTTGGGTGCTACTTCCCAACTGGCCGCTCTTAGGAACTGGTTAAACATTAACTCAGGTGTGACTCAGCGAGCCCCTGGGTCATGGCTATTCTCCTTGCTGCTCTGCTCTCCTAAGTCTGGTTTGAGG CTCCTCTCCCTCCCTGGCGTCCCAGC					
Parent sequence (SK-N-SH)	CGAAATTCCCACAACTGGAATATATACCGCATTGATACTATCTTCCCCTTTATCCTAAAACTGTCATCTTTAAGCTTCAAACAGTACTGTGATTTGATGTTAACCATGATGATTTGTACAACCGTATTTACAAGGCCACAACTCAAGTCATTTTTAGATTTCATCCTTCTAGAC TGTAGCCAAATTCCCATCCATAAAGC					
Edited sequence (SK-N-SH)	GGAAATTCCCACAACTGGAATATATACCGCATTGATACTATCTTCCCCTTTATCCTAAAACTGTCATCTTTAAGCTTCAAACAGTACTGTGATTTGATGTTAACCATGATGATTTGTACAACCGTATTTACAAGGCCACAACTCAAGCTATTTTTAGATTTCATCCTTCTGGAA TGTAGCCAAATTCCCATCCATAAAGC					

Note: P = parent natural CRE; E = SafeEdit-CRE edited design. Primary margin and target gain are from the Malinois predictor used to direct the search; reviewer margin and uncertainty are from the independent residual-dilated ensemble used during cross-model reranking. All edited sequences retain length 200 nt and exact edit budget. The full 74-candidate Tier A library is provided as a machine-readable table in the frozen release bundle.

3 Reproducibility summary

Parent selection was deterministic and label-blind, with all historical identifiers, sequence hashes, and audit-flagged conflicts removed before benchmark construction. The post-freeze evaluator was loaded only after candidate files and their SHA-256 manifests had been written. The compute-matched beam used the same beam width, expansion depth, and pre-screen size as SafeEdit; random comparisons averaged three fixed replicates. Final confidence intervals used 100,000 parent-cluster bootstrap replicates. All seven ablation configurations reran the complete search, and every reported figure and table can be regenerated from the candidate-level data included with the analysis scripts.